2.2 Group 2

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

When a firework is lit a fuel, and an oxidising agent reacts.

In such a firework, magnesium is the fuel, and barium nitrate is the oxidising agent.

Which solid products are produced when the firework is lit?

1 BaO

2 MgO

3 $Mg(NO_3)_2$

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 2

Group II nitrates undergo thermal decomposition according to the following equation.

$$X(NO_3)_2 \rightarrow XO + 2NO_2 + \frac{1}{2} O_2$$

Which Group II nitrate requires the highest temperature to bring about its thermal decomposition?

A. Barium nitrate

B. Calcium nitrate

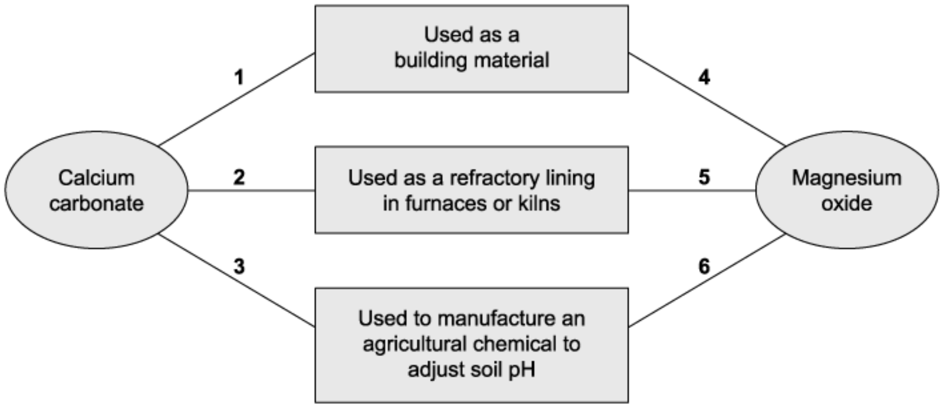
C. Magnesium nitrate

D. Strontium nitrate

[1 mark]

Question 3

The diagram shows some applications of compounds of Group II elements.



Which numbered links are correct?

	Magnesium oxide	Calcium carbonate
A	4 and 5 only	1, 2 and 3
B	4 and 6 only	1, 2 and 3
C	5 only	1 and 3 only
D	6 only	1 and 3 only

[1 mark]

Question 4

Which mass of solid residue will be obtained from the thermal decomposition of 5.10 g of anhydrous calcium nitrate?

Use of the periodic table is relevant to this question.

A. 0.70 g

B. 0.87 g

C. 1.40 g

D. 1.74 g

[1 mark]

Question 5

Thermal decomposition of magnesium nitrate, $Mg(NO_3)_2$, produces a white solid and a mixture of gases. One of the gases released is oxygen.

32.3 g of anhydrous magnesium nitrate is heated until no further reaction takes place.

Use of the periodic table is relevant to this question.

What is the mass of oxygen produced?

A. 8.32 g

B. 3.52 g

C. 12.48 g

D. 19.2 g

[1 mark]

Question 6

In agriculture, lime can be added to damp soil to change the pH. This can be followed by the addition of a nitrogenous fertiliser ammonium sulfate. This will result in a chemical reaction in the soil.

Which substances were formed in this reaction?

1 Sulfuric acid

2 Calcium sulfate

3 Ammonia

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 7

As you go down Group II, what are the trends in the properties in the table below?

	Decomposition temperature of the carbonate	First ionisation energy
A	Decreases	Decreases
B	Decreases	Increases
C	Increases	Increases
D	Increases	Decreases

[1 mark]

Question 8

When quicklime, water and sand are mixed, it makes lime mortar. Lime mortar hardens over time. Aqueous hydrochloric acid reacts with fresh and old lime mortar, but only the old lime mortar effervesces during the reaction.

Which equation describes the change from fresh to old lime mortar?

A. $CaO + CO_2 \rightarrow CaCO_3$

B. $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

C. $Ca(OH)_2 \rightarrow CaO + H_2O$

D. $CaO + H_2O \rightarrow Ca(OH)_2$

[1 mark]

Question 9

A compound was thermally decomposed to a constant mass; this gave off a colourless gas that reacted with limewater. The limewater went cloudy. The solid that was left was allowed to cool, and aqueous hydrochloric acid added, a vigorous effervescence was seen.

Use of the periodic table is relevant to this question.

What was the mineral compound?

A. Barytocalcite, $BaCO_3.CaCO_3$

B. Artinite, $MgCO_3.Mg(OH)_2.3H_2O$

C. Aragonite, $CaCO_3$

D. Dolomite, $CaCO_3.MgCO_3$

[1 mark]

Question 10

Thermal decomposition of compound **S** releases carbon dioxide gas and forms a white solid, **T**. **S** and **T** are not soluble in water. Compound **T** is used as a refractory kiln lining.

What is compound **S**?

A. $CaCO_3$

B. $MgCO_3$

C. CaO

D. MgO

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- **The equation for this reaction.**
  - $Ba(NO_3)_2 + 5Mg \rightarrow 5MgO + BaO + N_2$
- The barium nitrate is acting as an oxidising agent producing oxygen when heated; this leaves the barium oxide and nitrogen gas as the other products of this reaction.
- The oxygen produced reacts with the magnesium to produce magnesium oxide.

2

The correct answer is **A** because:

- Thermal decomposition of Group II nitrates requires higher temperatures as you go down the group.
- The smaller positive ions at the top of Group II will polarise the nitrate ions more than the larger ions at the bottom of the group.
- The more polarised they are, the more likely they are to thermally decompose.
- .Therefore, you have to heat the barium compound more strongly to thermally decompose the compound
  - The nitrates are more thermally stable as you go down the group.

3

The correct answer is **C** because:

- Magnesium oxide is used as kiln lining as it has a very high melting point.
- Calcium carbonate (limestone) is used in building materials.

- Calcium carbonate is used to improve the quality of soil and in manufacturing processes.
- The correct answer is option **C.**

4

The correct answer is **D** because:

- **The equation for this reaction is**
  - $Ca(NO_3)_2\ (s) \rightarrow CaO\ (s) + 2NO_2\ (g) + ½\ O_2\ (g)$
- **The ratio of the reaction**
  - 1 mole $Ca(NO_3)_2$ will produce 1 mol CaO
  - Molar mass $Ca(NO_3)_2$ = 164.0 g mol^{-1}
- **Number of moles of $Ca(NO_3)_2$**
  - moles= $\frac{mass}{M_r} = \frac{5.10}{164.0} = 0.031\ mole$
  - Therefore 0.031 moles of CaO is produced.
- **Mass of CaO**
  - mass = moles × M_r = 0.031 × 56 = 1.74 g

5

The correct answer is **B** because:

- **The equation for this reaction is**
  - $Mg(NO_3)_2\ (s) \rightarrow MgO\ (s) + ½\ O_2\ (g) + 2NO_2\ (g)$
  - to get 1 mole of oxygen you have to multiply everything by 2
  - $2Mg(NO_3)_2\ (s) \rightarrow 2MgO\ (s) + O_2\ (g) + 4NO_2\ (g)$
- **The ratio of the reaction**
  - 2 mole $Mg(NO_3)_2$ will produce 1 mol oxygen
  - Molar mass $Mg(NO_3)_2$ = 148.3 g mol^{-1}
- **Number of moles of $Mg(NO_3)_2$**
  - $moles = \frac{mass}{M_r} = \frac{32.3}{148.3} = 0.22\ mole$

- Therefore 0.11 moles of O_2 is produced

- **Mass of O_2**
    - mass = moles × M_r = 0.11 × 32 = 3.52 g

6

The correct answer is **C** because:

- Lime is calcium oxide or calcium hydroxide.

- The question states the soil is damp so providing conditions needed for the two solids to react.

- The chemical reaction that would take place is
    - $(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow 2NH_3 + 2H_2O + CaSO_4$

- Therefore, the two products made in this reaction would be ammonia and calcium sulfate.

7

The correct answer is **D** because:

- **Thermal decomposition of Group II carbonates requires higher temperatures as you go down the group.**

- The smaller positive ions at the top of Group II will polarise the carbonate ions more than the larger ions at the bottom of the group.

- The more polarised they are, the more likely they are to thermally decompose.

- Therefore, you have to heat the barium compound more strongly to thermally decompose the compound

- Therefore, the two products made in this reaction would be ammonia and calcium sulfate.

7

The correct answer is **D** because:

- **Thermal decomposition of Group II carbonates requires higher temperatures as you go down the group.**
- The smaller positive ions at the top of Group II will polarise the carbonate ions more than the larger ions at the bottom of the group.
- The more polarised they are, the more likely they are to thermally decompose.
- Therefore, you have to heat the barium compound more strongly to thermally decompose the compound.
- **As you go down the group the first ionisation energy of Group II decreases.**
- There are more filled shells between the nucleus and the outer electrons; this shields the outer electrons from the attraction of the nucleus.
- The radius of the atom increases, so the distance between the nucleus and the outer electrons increases.
  - Therefore, the force of attraction between the nucleus and outer electrons is reduced.
  - So, less energy is needed to remove an outer electron.

8

The correct answer is **B** because:

- **The equation for this reaction is**
  - $Ca(OH)_2 (aq) + CO_2 (aq) \rightarrow CaCO_3 (s) + H_2O (l)$
- As the calcium hydroxide in the fresh lime mortar reacts over time with the carbon dioxide in the air, it makes a solid calcium carbonate.
- Over time the water produced will evaporate leaving a much harder calcium

carbonate.

- The calcium carbonate will effervesce when added to acid due to the production of carbon dioxide gas.

- **The equation for this reaction is**
  - $CaCO_3\ (aq) + 2HCl\ (aq) \rightarrow CaCl_2\ (aq) + CO_2\ (g) + H_2O\ (l)$

A is incorrect as calcium oxide will react with carbon dioxide to produce calcium carbonate, but when mixed with water calcium oxide forms calcium hydroxide. So, the fresh quicklime will be calcium hydroxide, not calcium oxide.

C is incorrect as this is the reverse of the reaction of calcium oxide and water; this would require lots of energy, (heated to 512˚C), to take place as the forward reaction is exothermic.

D is incorrect as this is the reaction to make fresh lime mortar not the conversion from fresh to old lime mortar.

9

The correct answer is **A** because:

- When Group II metal carbonates are thermally decomposed they release carbon dioxide that would react with the lime water to turn it cloudy.

- The equation for this reaction is
  - $CaCO_3\ (s) \rightarrow CaO\ (s) + CO_2\ (g)$

- Thermal decomposition requires higher temperatures as you go down Group II.

- The smaller positive ions at the top of Group II will polarise the carbonate ions more than the larger ions at the bottom of the group.

- The more polarised they are, the more likely they are to thermally decompose.

- Therefore, you have to heat the barium compound more strongly to thermally decompose; this would not happen in a Bunsen burner.

- In order to produce a vigorous effervescence, a carbonate would need to be present to give of the CO_2 with the acid.

B is incorrect as the product from the heating would be magnesium oxide; this would produce salt and water with the addition of the acid.

C is incorrect as the product from the heating would be calcium oxide; this would produce salt and water with the addition of the acid.

D is incorrect as the product from the heating would be magnesium oxide and calcium oxide; this would produce salt and water with the addition of the acid.

10

The correct answer is **B** because:

- The question states that the compound produced from the thermal decomposition **(T)** is used as a refractory kiln lining.
- This will need to withstand very high temperatures. Magnesium oxide has the highest melting point so it would be used as kiln lining.
- Compound **S** must be magnesium carbonate as when Group II metal carbonates are thermally decomposed, they release carbon dioxide.
    - The white solid produced would be the metal oxide.

- Both magnesium carbonate (T) and magnesium oxide (S) are insoluble in water.

A is incorrect as calcium oxide produced from the thermal decomposition of calcium carbonate is soluble in water producing calcium hydroxide.

C & **D** are incorrect as to thermally decompose and produce carbon dioxide the compound would be a metal carbonate.